Network Troubleshooting Quiz — Round 2

Scenario-based troubleshooting with Cisco show commands, configurations, logs, and routing behavior

Instructions: Choose the single best answer. No answer key is included. Focus on the evidence in each scenario and identify the most likely root cause.

Layer 2 Troubleshooting

1. Access Port — Interface Up, No Connectivity

A PC is connected to Gi1/0/10. The interface is operational, but the PC cannot reach other hosts in its VLAN.
`show interfaces gi1/0/10 status` reports:
`Gi1/0/10 connected trunk a-full a-1000`
The PC should be in VLAN 20. What is the most likely problem?

A. The port is configured as a trunk instead of an access port

B. STP has elected the wrong root bridge

C. DHCP snooping is disabled

D. The switch has no default route

2. VLAN Missing From Trunk

A host in VLAN 30 can communicate locally but cannot reach VLAN 30 hosts connected to another switch. `show interfaces trunk` on the inter-switch link shows:
`Vlans allowed on trunk: 10,20,40`
`Vlans allowed and active: 10,20,40`
What is the best diagnosis?

A. VLAN 30 is not allowed on the trunk

B. VLAN 30 is configured as the native VLAN

C. The access ports need PortFast

D. OSPF is missing area 30

3. Native VLAN Mismatch

A Cisco switch logs:
`%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on Gi1/0/48`
Switch A uses native VLAN 99; Switch B uses native VLAN 1. What should you correct?

A. Configure the same native VLAN on both ends

B. Disable STP on the trunk

C. Configure both interfaces as access ports

D. Increase the STP priority

4. Voice VLAN

A phone and PC share Gi1/0/15. The PC receives an address in VLAN 10, but the phone receives no DHCP address. `show run interface gi1/0/15` shows:

```
switchport mode access
switchport access vlan 10
```

Which configuration is most likely required?

A. switchport voice vlan 20

B. switchport mode trunk

C. switchport trunk allowed vlan 20

D. no spanning-tree portfast

5. MAC Address Learning

A host at MAC 0011.2233.4455 is connected to Gi1/0/7. A technician runs:

```
show mac address-table dynamic vlan 20
```

No entry for 0011.2233.4455 appears. Which is the best next troubleshooting step?

A. Verify the physical port and whether frames are being received

B. Change the router's OSPF process ID

C. Change the switch's default gateway

D. Clear the BGP table

6. MAC Flapping

The switch repeatedly logs:

```
MACFLAP_NOTIF: Host aaaa.bbbb.cccc in vlan 20 is flapping between Gi1/0/1 and
Gi1/0/2
```

Both interfaces connect to the same access-layer area. What should you investigate first?

A. A Layer 2 loop or unintended redundant connection

B. DNS configuration

C. OSPF authentication

D. NAT overload

7. Port Security Violation

Gi1/0/8 is configured:

```
switchport port-security
switchport port-security maximum 1
switchport port-security violation shutdown
```

The interface is err-disabled after a different device is connected. Which command is most useful to confirm the cause?

A. show port-security interface gi1/0/8

B. show ip ospf neighbor

C. show ip bgp summary

D. show ip nat translations

8. BPDU Guard

An access port configured with PortFast and BPDU Guard becomes err-disabled immediately after a user connects a small switch. What evidence would confirm the diagnosis?

A. show spanning-tree interface gi1/0/10 detail showing BPDU Guard violation

B. show ip route showing a missing default route

C. show ip arp showing an incomplete entry

D. show ip bgp summary showing Active

9. STP Root Problem

You intended SW1 to be the STP root for VLAN 10. On SW2:
`show spanning-tree vlan 10`
shows SW2 as the root bridge. SW1 has priority 32768 and SW2 has priority 24576. What should you do on SW1?

A. Lower SW1's STP priority below SW2's

B. Increase SW1's STP priority

C. Enable BPDU Guard on the trunk

D. Disable VLAN 10

10. PortFast Mistake

A technician enabled PortFast on an interface connecting SW1 to SW2. What is the primary concern?

A. PortFast is intended for edge/access ports and can be unsafe on switch-to-switch links

B. PortFast prevents all VLAN traffic

C. PortFast disables MAC learning

D. PortFast converts an access port to a trunk

11. EtherChannel Failure

Two links between switches are intended to form an EtherChannel. On SW1:
`show etherchannel summary`
shows Gi1/0/1 as (P) but Gi1/0/2 as (s). What does this indicate?

A. One link is bundled while the other is suspended

B. Both links are forwarding normally in the bundle

C. Both links are Layer 3 routed ports

D. STP has disabled the entire switch

12. EtherChannel Load Balancing

An EtherChannel has four physical links, but almost all traffic appears to use one member. The bundle is healthy and all links show (P). What should you check?

A. The EtherChannel load-balancing method and traffic flow characteristics

B. The STP root password

C. DHCP snooping database

D. Native VLAN mismatch only

13. VLAN Pruning

VLAN 50 works between Switch A and B but stops at Switch C. The trunks are operational. A VLAN pruning configuration prevents VLAN 50 from being forwarded across one trunk. What should you verify?

A. Whether VLAN 50 is being pruned/filtered on the relevant trunk

B. Whether PortFast is enabled

C. Whether the router has a default route

D. Whether BGP MED is correct

14. HSRP Failover

R1 and R2 provide gateway redundancy. R1 has HSRP priority 110; R2 has priority 100. R1 fails, and R2 becomes active. When R1 returns, R2 remains active even though R1 has higher priority. What should you check?

A. Whether HSRP preemption is configured

B. Whether OSPF authentication is enabled

C. Whether DHCP snooping is disabled

D. Whether VLAN 1 is native

15. DHCP Snooping

Clients stop receiving DHCP addresses after DHCP snooping is enabled. The DHCP server is connected through Gi1/0/24. Which configuration issue is most likely?

A. The DHCP-server-facing interface is not trusted

B. PortFast is disabled on the server

C. The switch needs a lower STP priority

D. The DHCP server must be in the native VLAN

16. Dynamic ARP Inspection

A client can obtain a DHCP address, but its ARP packets are being dropped after Dynamic ARP Inspection is enabled. Which dependency should you investigate?

A. DHCP snooping bindings and DAI trust/configuration

B. BGP route reflectors

C. OSPF router ID

D. EtherChannel hashing

17. Root Guard

A downstream switch port becomes root-inconsistent after a BPDU is received. The port is configured with Root Guard. What does this tell you?

A. The port received a superior BPDU and Root Guard is preventing it from becoming a path toward the root

B. DHCP snooping has blocked the port

C. The interface has a physical duplex mismatch

D. NAT has failed

18. Switch ACL

Users in VLAN 20 can reach their default gateway but cannot reach a server subnet. A switch SVI has an inbound ACL applied. Which command should you use first to inspect whether the ACL is blocking traffic?

A. show access-lists

B. show spanning-tree root

C. show etherchannel summary

D. show mac address-table dynamic

Layer 3 Routing Troubleshooting

19. Static Route / Wrong Next Hop

A router has:
`S 172.16.50.0/24 [1/0] via 10.0.0.2`
Traffic to 172.16.50.0/24 fails. The actual neighboring router that leads to the destination is 10.0.0.3. What is the most likely issue?

A. The static route points to the wrong next hop

B. The route has an administrative distance that is too low

C. ARP is unnecessary for a static route

D. The destination must be a directly connected network

20. Route Exists, Traffic Fails

The routing table contains a valid route to 10.10.20.0/24 via 192.168.1.2. A ping fails. `ping 192.168.1.2` also fails. What should you investigate first?

A. Reachability of the next hop and the underlying link

B. BGP best-path selection

C. STP root priority

D. VLAN pruning on unrelated interfaces

21. Asymmetric Routing

A firewall permits outbound sessions but drops return traffic because the return path enters through a different interface. The network diagram shows different forward and reverse paths. What problem is most likely?

A. Asymmetric routing

B. Native VLAN mismatch

C. MAC learning failure

D. OSPF area mismatch

22. ARP Failure

A router considers 192.168.10.50 directly connected, but `show arp 192.168.10.50` shows an incomplete entry. What is the immediate problem?

A. The router cannot resolve the destination MAC address through ARP

B. The router has no route to the subnet

C. BGP has withdrawn the route

D. STP has selected the wrong root

23. MTU Mismatch

A large ping with the DF bit set fails between two routers, while small pings work. The path contains a link with a smaller MTU. What should you investigate?

A. MTU mismatch/path MTU and fragmentation behavior

B. HSRP priority

C. VLAN pruning

D. Port security

24. VRF Misconfiguration

A router has two VRFs. A route to 10.20.20.0/24 exists in VRF BLUE, but a host in VRF RED cannot reach it. What should you verify?

A. VRF membership and the routing table associated with the source interface

B. STP PortFast

C. Native VLAN

D. BGP MED only

25. WAN Link Up, Traffic Down

A WAN interface reports up/up, but no traffic passes. The carrier confirms the circuit is active. The router has no route toward the remote subnet. What is the most likely issue?

A. Missing/incorrect routing information

B. BPDU Guard

C. Voice VLAN failure

D. EtherChannel hashing

OSPF & BGP Troubleshooting

26. OSPF INIT

R1 shows R2's OSPF neighbor stuck in INIT. R1 receives R2's Hello packets, but R2's Hello does not list R1 as a neighbor. What should you investigate?

A. Whether R2 is receiving R1's Hellos correctly

B. Whether BGP has a route reflector

C. Whether DHCP snooping is trusted

D. Whether STP priority is 0

27. OSPF EXSTART

Two OSPF routers reach EXSTART but never progress. The interfaces have different MTU values. What is the most likely cause?

A. MTU mismatch preventing database exchange

B. Wrong HSRP priority

C. Missing voice VLAN

D. DHCP snooping

28. OSPF Area/Network Type

Two routers share a subnet, but one interface is configured for OSPF area 0 and the other for area 1. They also use different OSPF network types. What is the likely result?

A. OSPF adjacency may fail because required parameters do not match

B. BGP automatically forms a session

C. The switch creates a new VLAN

D. NAT fixes the mismatch

29. OSPF Routes Not Installing

OSPF neighbors are FULL and LSAs are present. A route appears in the OSPF database but not in the routing table. What should you investigate?

A. Whether the route is the best available path and whether its next hop is usable

B. Whether PortFast is enabled

C. Whether the access port has a voice VLAN

D. Whether the switch learned the server MAC

30. BGP Established, No Routes

BGP shows:
`Neighbor 203.0.113.2 State/PfxRcd Established/0`
The session is established, but zero prefixes are received. What should you check first?

A. Whether the peer is advertising prefixes and whether inbound policy permits them

B. STP root bridge priority

C. DHCP snooping bindings

D. EtherChannel hashing

31. BGP Prefix Not Advertised

R1 has 10.50.50.0/24 in its routing table. Its BGP neighbor does not receive the prefix. The BGP configuration does not contain a network statement or redistribution for 10.50.50.0/24. What is the likely cause?

A. The prefix is not being injected/advertised into BGP

B. The BGP session must be reset because all routes are automatic

C. STP is blocking the prefix

D. ARP must be configured for BGP

32. BGP Route Flapping

A BGP prefix repeatedly appears and disappears. Interface logs show the physical WAN interface repeatedly going down and up. What should you investigate first?

A. The underlying physical/WAN instability

B. BGP local preference only

C. VLAN pruning

D. DHCP snooping

33. BGP Path Selection

Two BGP paths exist for the same prefix. Path A has local preference 200; Path B has local preference 100. Assuming other relevant factors do not override it, which path should be preferred?

A. Path A

B. Path B

C. The path with the higher MED always

D. The path learned later

34. Prefix Accepted but Not Installed

A BGP prefix appears in the BGP table but not the IP routing table. Another route to the same prefix exists from OSPF with a lower administrative distance. What is the likely explanation?

A. The OSPF route is preferred for installation because of administrative distance

B. BGP cannot install any routes learned from eBGP

C. STP has blocked the route

D. ARP must be disabled

35. Blackhole Route

A route to 10.99.99.0/24 points to a Null0 interface. Traffic to that destination is silently discarded. What type of configuration could cause this?

A. An intentional or accidental blackhole/null route

B. PortFast

C. DHCP snooping

D. Native VLAN mismatch